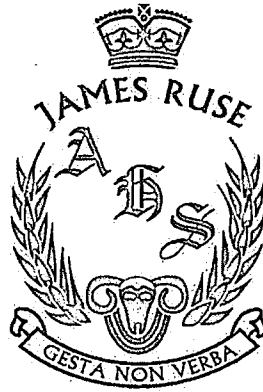


Name:	
Class:	



YEAR 12

TRIAL
2016

MATHEMATICS EXTENSION 1

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 2 hours.
- Write in black or blue pen.
- Board approved calculators & templates may be used
- A Reference Sheet is provided.
- In Question 11 - 14, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged working.

Total Marks 70

Section I: 10 marks

- Attempt Question 1 – 10.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II: 60 Marks

- Attempt Question 11 – 14
- Answer on blank paper unless otherwise instructed. Start a new page for each new question.
- Allow about 1 hour & 45 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labelled Question 11, Question 12, etc. Each question must show your Candidate Number.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1. Let α , β and γ be the roots of $x^3 + px^2 + q = 0$.

Express $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}$ in terms of p and q .

(A) pq

(B) $-pq$

(C) $-\frac{p}{q}$

(D) $\frac{p}{q}$

2. If $f(x) = \frac{3 + e^{2x}}{4}$, what is $f^{-1}(x) = ?$

(A) $\ln(4x - 3)$

(B) $\frac{1}{2}\ln(4x - 3)$

(C) $\ln 4x - \ln 3$

(D) $\frac{1}{2}(\ln 4x - \ln 3)$

3. A family of 6 adults and 3 children are randomly sitting around a circular table.

What is the number of possible seating arrangements if none of the children sit together?

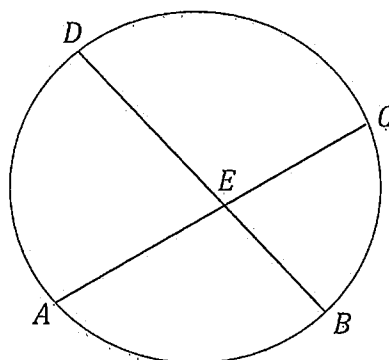
(A) 720

(B) 2 400

(C) 14 400

(D) 86 400

4. In the diagram below, chords AC and BD of the circle intersect at E .
 $AC = 9$ units, $CE = a$ units, $DE = x$ units and $EB = y$ units



NOT TO
SCALE

Which of the following statements is true?

- (A) $xy = 9a$
- (B) $\frac{x}{y} = \frac{9-a}{a}$
- (C) $x(x+y) = a(9-a)$
- (D) $xy = a(9-a)$
5. For $0 < x < 1$, what is $\frac{d}{dx} \left(\cos^{-1} \frac{1}{x} \right) = ?$

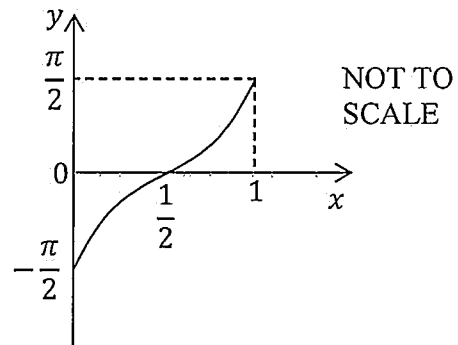
- (A) $\frac{-1}{x\sqrt{x^2-1}}$
- (B) $\frac{1}{x\sqrt{x^2-1}}$
- (C) $\frac{-1}{\sqrt{x^2-1}}$
- (D) $\frac{-x}{\sqrt{x^2-1}}$

6. What is the general solution of $\cos 2\theta = \frac{\sqrt{2}}{2}$?

- (A) $\theta = \frac{\pi}{8} + n\pi$ or $\theta = \frac{7\pi}{8} + n\pi$, n is an integer
- (B) $\theta = \frac{\pi}{8} + 2n\pi$ or $\theta = \frac{7\pi}{8} + 2n\pi$, n is an integer
- (C) $\theta = \frac{2\pi}{3} + n\pi$ or $\theta = \frac{4\pi}{3} + n\pi$, n is an integer
- (D) $\theta = \frac{\pi}{4} + n\pi$ or $\theta = \frac{3\pi}{4} + n\pi$, n is an integer

7. Which integral is obtained when the substitution $x = \sin \theta$ is applied to $\int \frac{x^2}{\sqrt{1-x^2}} dx$?
- (A) $\int \cos^2 \theta d\theta$
- (B) $\int \tan^2 \theta d\theta$
- (C) $\int \sin^2 \theta d\theta$
- (D) $\int \sin^2 \theta \cos^2 \theta d\theta$
8. A particle is moving in simple harmonic motion with its velocity given by $v^2 = 16(9 - x^2)$, where x is the displacement.
- What is the amplitude A , and the period T , of the motion?
- (A) $A = 3$ and $T = \frac{\pi}{4}$
- (B) $A = 3$ and $T = \frac{\pi}{2}$
- (C) $A = 4$ and $T = \frac{\pi}{4}$
- (D) $A = 4$ and $T = \frac{\pi}{2}$
9. What is the term independent of x in the expansion of $\left(x^3 + \frac{2}{x}\right)^{20}$?
- (A) $\binom{20}{10} 2^{10}$
- (B) $\binom{20}{5} 2^{15}$
- (C) $\binom{20}{4} 2^{16}$
- (D) $\binom{20}{5} 2^{25}$

10. The diagram shows the graph of a function.



Which function does the graph represent?

- (A) $y = -\cos^{-1}(2x - 1)$
- (B) $y = \sin^{-1}(2x - 1)$
- (C) $y = \sin^{-1}(x - 1)$
- (D) $y = -\cos^{-1}(x - 1)$

Section II

60 Marks

Attempt questions 11-14

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks) Use a SEPARATE writing booklet.

- (a) Let $P(x) = x^3 - ax^2 + x$ be a polynomial, where a is a real number. 2
When $P(x)$ is divided by $x - 3$ the remainder is 12.

Express $P(x)$ as a product of its factors.

- (b) Solve $\frac{5}{x-4} \geq 1$ 2

- (c) Find the value of 2

$$\lim_{x \rightarrow 0} \frac{3x \cos 4x}{\sin 2x}$$

Show all necessary working.

- (d) Use the substitution $u = 1 + \sqrt{2x}$ to find 3

$$\int_{\frac{1}{2}}^2 \frac{1}{1 + \sqrt{2x}} dx$$

- (e) Find the exact value of $\cos(2 \cos^{-1} x + \sin^{-1} x)$ when $x = \frac{1}{5}$. 3

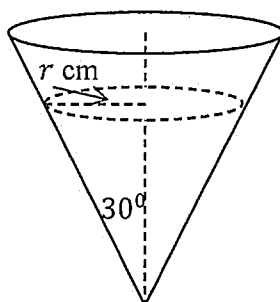
- (f) Find $\frac{d}{dx} \left(\tan^{-1} \frac{x}{3} \right)^2$ and hence find the exact value of 3

$$\int_0^{\sqrt{3}} \frac{\tan^{-1} \frac{x}{3}}{x^2 + 9} dx$$

End of Question 11

QUESTION 12 (15 Marks) Use a SEPARATE writing booklet.

- (a) In the figure below the angle between the axis and the slant edge of the right cone is 30° . Water is poured into the cone at a constant rate of 2 cm^3 per second. At time t seconds, the radius of the water surface is $r \text{ cm}$ and the volume of water in the cone is $V \text{ cm}^3$.

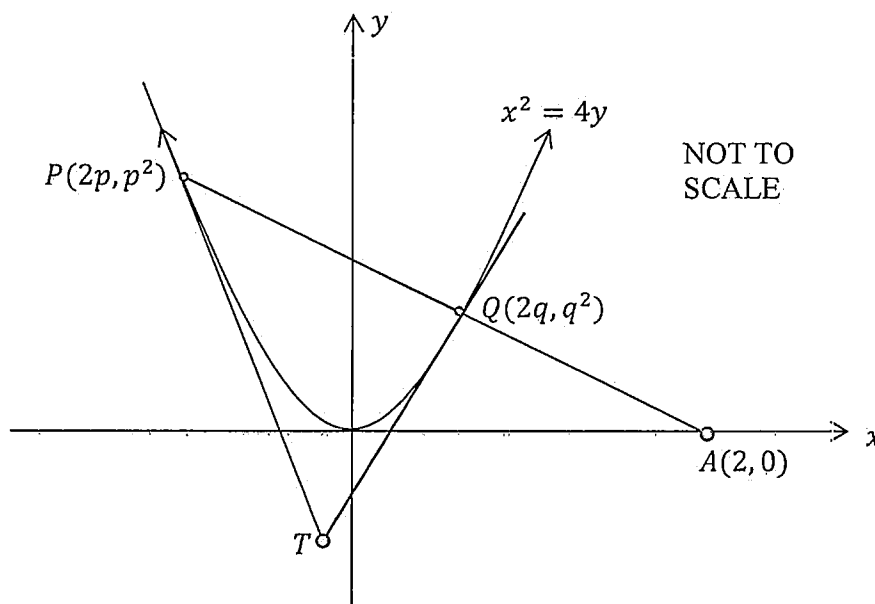


- (i) Show that $V = \frac{\sqrt{3}}{3} \pi r^3$. 1
- (ii) Hence, find the rate at which r is changing when $r = 2 \text{ cm}$. 2
- (b) The velocity of a particle moving in a straight line is given by
- $$v^2 = -9x^2 + 18x + 27.$$
- (i) Prove that the motion is simple harmonic. 2
- (ii) Hence find the centre of the motion. 1
- (iii) Find the maximum acceleration of the particle and state where it occurs. 2
- (c) Taking $x = -\frac{\pi}{6}$ as a first approximation to the root of the equation $2x + \cos x = 0$, 2
use Newton's method once to show that a second approximation to the root of the equation is $\frac{-\pi - 6\sqrt{3}}{30}$.
- (d) An unbiased die is thrown 6 times. Calculate the probabilities that the six scores obtained
- (i) will have a product which is an even number 2
- (ii) will consist of exactly two 6's and four odd numbers 3

End of Question 12

QUESTION 13 (15 Marks) Use a SEPARATE writing booklet.

- (a) Let $P(2p, p^2)$ and $Q(2q, q^2)$ be two points on the parabola $x^2 = 4y$. The tangents to the parabola at P and Q intersect at T . The secant PQ passes through the point $A(2, 0)$.



- (i) Show that $p + q = pq$. 2
- (ii) Assuming that the equation of the tangents at P and Q are $y = px - p^2$ and $y = qx - q^2$ respectively, determine the cartesian equation of the locus of T , including any restrictions. 3
- (b) The region enclosed by the curve $y = \sin^{-1} x$ and the y -axis between $x = 0$ and $x = \frac{\sqrt{3}}{2}$ is rotated about the y -axis to form a solid. 3
- Find the exact volume of the solid of revolution formed.
- (c) A manufacturer of metal pistons finds that on the average, 12% of the pistons are rejected because they are either oversize or undersize.
- What is the probability that a batch of 10 pistons will contain no more than 2 rejects? 3
- Express your answer to three decimal places.

Question 13 continues on page 9

Question 13 (continued)

- (d) A household iron is cooling in a room where the temperature of the air is constant at 22°C .

After t minutes the temperature T of the iron is modelled by the equation
 $T = 22 + Ae^{-kt}$, where A and k are constants.

The initial temperature of the iron is 80°C and it cools to 60°C after 10 minutes.

- (i) Find the exact values of the constants A and k . 2
- (ii) How long will it take for the temperature of the iron to cool to 30°C ? 2
Give your answer to the nearest minute.

End of Question 13

QUESTION 14 (15 Marks) Use a SEPARATE writing booklet.

- (a) Use mathematical induction to prove that for all positive integers n , 3

$$\sum_{r=1}^n r(r!) = (n+1)! - 1$$

- (b) The velocity of a particle moving in a straight line is given by $\dot{x} = e^x + e^{-x}$, where x in metres is the displacement from the origin. Initially the particle is at the origin.

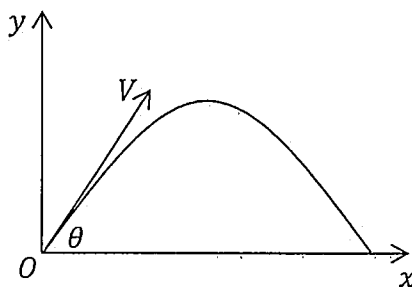
- (i) Show that the time taken by the particle as a function of its displacement is given by 2

$$t = \int \frac{e^x}{e^{2x} + 1} dx$$

- (ii) Hence, by using the substitution $u = e^x$, prove that 2

$$x = \ln \left(\tan \left(t + \frac{\pi}{4} \right) \right).$$

- (c) A particle P is fired from the ground at $t = 0$ with initial velocity V from the origin, at an angle of θ to the horizontal.



Assume that the horizontal and vertical equations of motion for the particle is given by

$$x_P = Vt \cos \theta \quad \text{and}$$

$$y_P = Vt \sin \theta - \frac{1}{2}gt^2 \quad \text{(DO NOT PROVE THESE)}$$

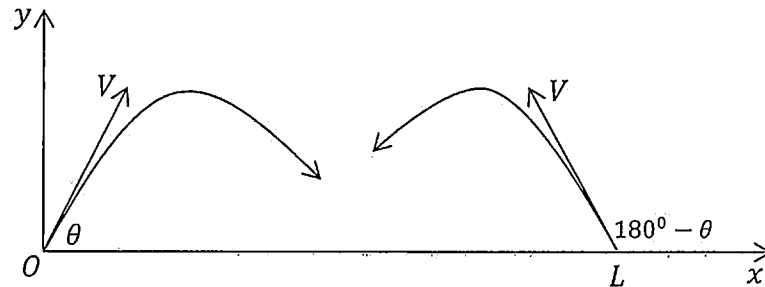
- (i) Show that the horizontal range of the projectile, R_P , is given by 3

$$R_P = \frac{V^2 \sin 2\theta}{g}$$

Question 14 (c) continues on page 11

Question 14 (c) (continued)

A second particle Q , is fired back towards the origin with velocity V from the ground at a distance of L metres to the **right** of the origin at time $t = 0$, with an angle of $(180 - \theta)^\circ$ to the positive direction of the x -axis.



- (ii) Show that if the particles collide, it will occur when

2

$$t = \frac{L}{2V \cos \theta}$$

- (iii) For the particles to collide, it must occur while the particles are still in flight (i.e. above the ground).

3

Prove that, for the particles to collide in the air,

$$0 < L < \frac{4V^2 \cos \theta \sin \theta}{g}$$

End of paper

JRAHS : MATHEMATICS EXTENSION : TRIAL 2016

MCQ :

1. $x^3 + px + q = 0$

$$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$$

$$= \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$$

$$= \frac{-p}{-q}$$

$$= p/q \quad \text{D}$$

2. $x = \frac{3 + e^{2y}}{4}$

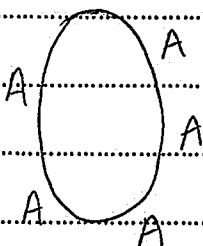
$$e^{2y} = 4x - 3$$

$$2y = \ln(4x - 3)$$

$$y = \frac{1}{2} \ln(4x - 3)$$

$\therefore$ B

3. A - fixed



The 6 adults can be arranged in $5!$ ways

If none of the children sit together, they would need to occupy 3 of the 6 positions (between 2 adults)

Hence 6C_3 ways.

But the 3 children can be arranged in $3!$ ways
Hence

$$5! \times {}^6C_3 \times 3!$$

But $5! \times 3! = 6!$

$$\therefore {}^6C_3 \times 6! \text{ ways} = 14400$$

$\therefore$ C

4. $DE/EB = EC/EA$

$$x/y = a/EA$$

$$\therefore AE = ay/x$$

But $DE \cdot EB = AE \cdot EC$

$$x \cdot y = (AC - EC) \cdot EC$$

$$= (9 - a) a$$

$$= a(9 - a)$$

$\therefore$ D

$$5. \frac{d}{dx} \left(\cos^{-1} \frac{1}{x} \right)$$

$$= - \frac{1}{\sqrt{1 - \frac{1}{x^2}}} \cdot \left(-\frac{1}{x^2} \right)$$

$$= \frac{1}{\frac{1}{x} \sqrt{x^2 - 1}} \cdot \frac{1}{x^2}$$

$$= \frac{1}{x \sqrt{x^2 - 1}}$$

$$\therefore \textcircled{B}$$

$$6. \cos 2\theta = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$\therefore 2\theta = \frac{\pi}{4} + 2n\pi$$

$$\therefore \theta = \frac{\pi}{8} + n\pi \text{ or}$$

$$2\theta = \frac{7\pi}{4} + 2n\pi$$

$$\therefore \theta = \frac{7\pi}{8} + n\pi$$

Hence $\textcircled{A}$

$$7. x = \sin \theta$$

$$dx = \cos \theta d\theta$$

$$\therefore I = \int \frac{\sin^2 \theta \cdot \cos \theta d\theta}{\sqrt{1 - \sin^2 \theta}}$$

$$= \int \frac{\sin^2 \theta \cdot \cos \theta \cdot d\theta}{\cos \theta}$$

$$= \int \sin^2 \theta d\theta$$

$$\therefore \textcircled{C}$$

$$8. v^2 = 16(9 - x^2) \\ = n^2(A^2 - x^2)$$

$$\therefore n = 4 \text{ and } A = 3$$

$$\Rightarrow T = 2\pi/4 = \pi/2$$

Hence Amplitude = 3
Period = $\pi/2$

$$\therefore \textcircled{B}$$

$$\begin{aligned}
 9. \quad & \left(x^3 + \frac{2}{x}\right)^{20} \\
 &= \sum_{r=1}^{20} {}^{20}C_r (x^3)^{20-r} \left(\frac{2}{x}\right)^r \\
 &= \sum {}^{20}C_r x^{60-3r} x^{-r} 2^r \\
 &= \sum {}^{20}C_r x^{60-4r} 2^r
 \end{aligned}$$

For term independent of x

$$60 - 4r = 0$$

$$\therefore r = 15$$

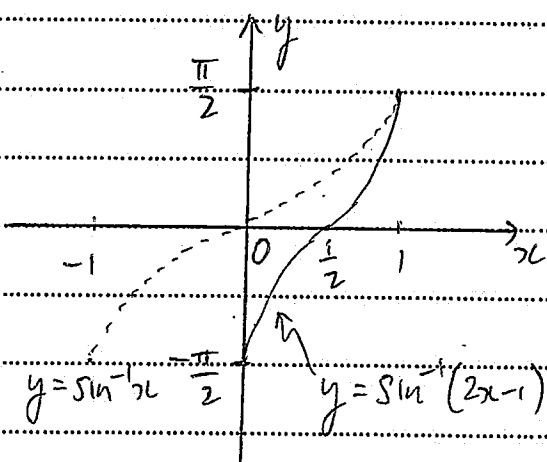
Hence the term is

$${}^{20}C_{15} x^0 \cdot 2^{15}$$

$$= {}^{20}C_5 2^{15}$$

$\therefore$ (B)

$$10. \quad y = \sin^{-1} x$$



$\therefore$ (B)

1. D

2. B

3. C

4. D

5. B

6. A

7. C

8. B

9. B

10. B

MATHEMATICS Extension 1 : Question...!!

Suggested Solutions	Marks	Marker's Comments
11(a) $p(x) = x^3 - ax^2 + x$ $p(3) = 12$ $\therefore 12 = 27 - 9a + 3$ $30 - 9a = 12$ $18 = 9a$ $a = 2$ $\therefore p(x) = x^3 - 2x^2 + x$ $= x(x^2 - 2x + 1)$ $= (x)(x-1)^2$	① ①	a lot of students wasted time using the factor theorem and long division.
(b) $\frac{5}{x-4} \geq 1$ $5(x-4) \geq (x-4)^2$ $x \neq 4$ $5(x-4) - (x-4)^2 \geq 0$ $(x-4)[5 - (x-4)] \geq 0$ $(x-4)(9-x) \geq 0$ $(x-4)(x-9) \leq 0$ $4 < x \leq 9$	① ①	*had to factorise correctly to get the first mark. *most students forget that $x \neq 4$.
(c) $\lim_{x \rightarrow 0} \frac{3x}{\sin 2x} \times \lim_{x \rightarrow 0} \cos 4x$ $= \lim_{x \rightarrow 0} \frac{2x}{\sin 2x} \times \frac{3}{2} \times \lim_{x \rightarrow 0} \cos 4x$ $= 1 \times \frac{3}{2} \times 1$ $= \frac{3}{2}$	① ①	*A lot of students used double angles in *lost a mark if you "lost" $\lim_{x \rightarrow 0}$!!!
(d) when $x = \frac{1}{2}$ $u = 2$ when $x = 2$ $u = 3$ $u = 1 + \sqrt{2x}$ $\therefore u = 1 + (2x)^{1/2}$ $\frac{du}{dx} = \frac{1}{2}(2x)^{-1/2} \times 2$ $\therefore \frac{du}{dx} = \frac{1}{\sqrt{2x}}$ $dx = \sqrt{2x} du$ $= (u-1) du$ (as $u = 1 + \sqrt{2x}$) $\therefore I = \int_2^3 \frac{1}{u} (u-1) du$ $= \int_2^3 (1 - \frac{1}{u}) du$ $= [u - \ln u]_2^3$	} ① ①	both limits correct

MATHEMATICS Extension 1 : Question 11. cont.

Suggested Solutions

Marks

Marker's Comments

$$= 3 - \ln 3 - (2 - \ln 2)$$

$$= 1 - \ln 3 + \ln 2$$

$$= 1 + \ln^2 \frac{2}{3} \quad \text{or} \quad 1 - \ln^{3/2}$$

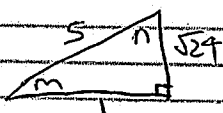
①

(e) $\cos(2\cos^{-1}x + \sin^{-1}x)$ now $x = \frac{1}{5}$

$$\cos(2\cos^{-1}\frac{1}{5} + \sin^{-1}\frac{1}{5})$$

let $\cos^{-1}\frac{1}{5} = m$ and $\sin^{-1}\frac{1}{5} = n$

$$\cos m = \frac{1}{5} \quad 0 < m < \frac{\pi}{2} \quad \sin n = \frac{1}{5} \quad 0 < n < \frac{\pi}{2}$$



$$\therefore \cos(2m + n)$$

①

$$= \cos 2m \cos n - \sin 2m \sin n$$

$$= (\cos^2 m - \sin^2 m) \cos n - 2 \sin m \cos m \sin n$$

①

$$= \left[\left(\frac{1}{5}\right)^2 - \left(\frac{\sqrt{24}}{5}\right)^2\right] \frac{\sqrt{24}}{5} - 2 \times \frac{\sqrt{24}}{5} \times \frac{1}{5} \times \frac{1}{5}$$

$$= \frac{-23}{25} \times \frac{\sqrt{24}}{5} - \frac{2\sqrt{24}}{125}$$

$$= \frac{-25\sqrt{24}}{125}$$

$$= -\frac{\sqrt{24}}{5}$$

①

$$= -\frac{2\sqrt{6}}{5}$$

(f) let $y = (\tan^{-1} \frac{x}{3})^2$

$$\frac{dy}{dx} = 2 \left(\tan^{-1} \frac{x}{3} \right)^1 \times \frac{3}{3^2 + x^2}$$

$$= \frac{6 \tan^{-1} \frac{x}{3}}{9 + x^2}$$

①

 $\int_0^{\sqrt{3}}$

$$\int_0^{\sqrt{3}} \frac{\tan^{-1} \frac{x}{3}}{9 + x^2} dx = \frac{1}{6} \left[\left(\tan^{-1} \frac{x}{3} \right)^2 \right]_0^{\sqrt{3}}$$

①

$$= \frac{1}{6} \left(\tan^{-1} \frac{\sqrt{3}}{3} \right)^2 - \left(\tan^{-1} 0 \right)^2$$

$$= \frac{1}{6} \left(\tan^{-1} \frac{1}{\sqrt{3}} \right)^2 - 0$$

$$= \frac{1}{6} \times \left(\frac{\pi}{6} \right)^2$$

$$= \frac{\pi^2}{216}$$

①

a lot of students forgot the "3" or had "9".

* Some students forgot to square the $\tan^{-1} \frac{x}{3}$

* said that some students didn't know their exact values.

MATHEMATICS: Question.....12
Extension 2 |

Suggested Solutions	Marks	Marker's Comments
(a) (i) $V = \frac{\pi r^2 h}{3}$ $= \frac{\pi r^2 (\sqrt{3} r)}{3}$ $= \frac{\sqrt{3} \pi r^3}{3}$	1	
(ii) $\frac{dr}{dt} = \frac{dV}{dt} \times \frac{dr}{dV}$ $= \frac{dV}{dt} \div \frac{dV}{dr}$ $= \frac{2}{\frac{\sqrt{3} \times 2 \pi r^2}{3}}$ $= \frac{2}{\sqrt{3} \pi r^2}$	1	
$r = 2 \Rightarrow \frac{dr}{dt} = \frac{2}{4\sqrt{3} \pi}$ $= \frac{1}{2\sqrt{3} \pi} \text{ cm s}^{-1}$ $= \frac{\sqrt{3}}{6\pi} \text{ cm s}^{-1}$	1	

MATHEMATICS: Question.....
Extension 2

Suggested Solutions

Marks

Marker's Comments

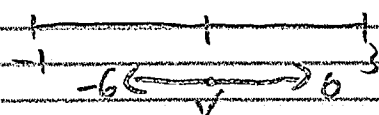
(b)

$$\begin{aligned} \text{(i)} \quad v^2 &= -9x^2 + 18x + 27 \\ &= -9(x^2 - 2x - 3) \\ &= -9(x-1)^2 - 4 \\ &= 9(4 - (x-1)^2) \\ &= 9^2(a^2 - (x-1)^2) \end{aligned}$$

$\therefore$ Motion is simple harmonic

(ii) x is a maximum when v is 0.

$$\begin{aligned} 0 &= (x+1)(x-3) \quad \text{from (i)} \\ x &= -1, 3 \end{aligned}$$



centre is $x=1$

(iii) acceleration is a maximum at the extremities, $x=3$, $x=-1$.

$$\begin{aligned} a &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ &= \frac{d}{dx} \left(\frac{-9}{2} x^2 + 9x + \frac{27}{2} \right) \\ &= -9x + 9 \end{aligned}$$

when $x=3$, $a=-18$

or $x=-1$, $a=18$

either solution
earns the mark.
BOTH a and
 x must be
stated together.

MATHEMATICS: Question.....
Extension 2

Suggested Solutions

Marks

Marker's Comments

$$(c) \quad 2x + \cos x = 0$$

$$\text{Let } f(x) = 2x + \cos x$$

$$f\left(-\frac{\pi}{6}\right) = -\frac{\pi}{3} + \cos\left(-\frac{\pi}{6}\right)$$

$$= -\frac{\pi}{3} + \cos\left(\frac{\pi}{6}\right)$$

$$= -\frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

$$f'(x) = 2 - \sin x$$

$$f'\left(-\frac{\pi}{6}\right) = 2 - \sin\left(-\frac{\pi}{6}\right)$$

$$= 2 + \frac{1}{2}$$

$$= \frac{5}{2}$$

Second application:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= -\frac{\pi}{6} - \left(\frac{-\frac{\pi}{3} + \frac{\sqrt{3}}{2}}{\frac{5}{2}} \right)$$

$$= -\frac{\pi}{6} + \frac{2\pi}{15} - \frac{\sqrt{3}}{5}$$

$$= \frac{-\pi - 6\sqrt{3}}{30}$$

As required

MATHEMATICS: Question.....
Extension 2

Suggested Solutions

Marks

Marker's Comments

(d) (i)

$$\begin{aligned} P(\text{even product}) &= 1 - P(\text{all odd}) \\ &= 1 - \left(\frac{1}{2}\right)^6 \\ &= \frac{63}{64} \end{aligned}$$

1

1

$$\begin{aligned} (ii) P(4 \text{ odds}) &= \frac{3}{6} \times \frac{3}{6} \times \frac{3}{6} \times \frac{3}{6} \\ &= \left(\frac{1}{2}\right)^4 \end{aligned}$$

$$P(2 \text{ 6s}) = \left(\frac{1}{6}\right)^2$$

1

$$P = {}^nC_r p^r q^{n-r}$$

$$= {}^6C_2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{2}\right)^{6-2}$$

1

$$= 15 \times \left(\frac{1}{6}\right)^2 \times \left(\frac{1}{2}\right)^4$$

1

$$= \frac{5}{192}$$

Alternative: Let odds be A, evens be B.

$$\text{Case 1: } (6 \text{ 6 A A A A}) = \frac{{}^6C_1 \times 6!}{2! \times 4!} = 45$$

$$\text{Case 2: } (6 \text{ 6 A A B B}) = \frac{{}^6C_2 \times 6!}{2! \times 2! \times 2!}$$

1

One mark

for being 500

the right track,

looking for correct

cases

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned}
 a) \quad m &= \frac{p^2 - q^2}{2p - 2q} \\
 &= \frac{(p-q)(p+q)}{2(p-q)} \\
 &= \frac{p+q}{2}
 \end{aligned}$$

1

 well
done

$$y - p^2 = \frac{1}{2}(p+q)(2x - 2p)$$

 goes through $A(2, 0)$

$$0 = \frac{1}{2}(p+q)2 - pq$$

$$\therefore p+q = pq$$

1

$$b) \quad y = px - p^2 \quad - (1)$$

$$y = qx - q^2 \quad - (2)$$

$$(1) - (2) \quad (p-q)x - (p^2 - q^2) = 0$$

$$(p-q)x = (p-q)(p+q)$$

$$\therefore x = p+q \quad p \neq q$$

$$\therefore y = q(p+q) \text{ sub in 2}$$

$$\begin{aligned}
 y &= pq - q^2 + q^2 \\
 &= \underline{pq}
 \end{aligned}$$

$$\therefore T(p+q, pq)$$

1

 students
who substituted
 $p+q = pq$ into
x then had
difficulty
finding y

Suggested Solutions

Marks

Marker's Comments

Now $p+q = pq$

$\therefore x = p+q \quad y = pq$

so $y = x$ is locus

Tangents meet outside of parabola

sub $y = x$ into $x^2 = 4y$

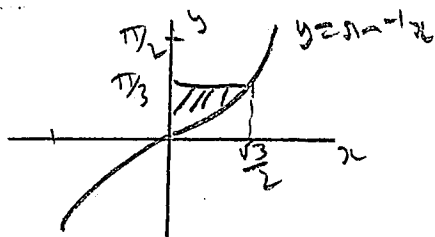
$$x^2 = 4x$$

$$x(x-4) = 0$$

$\therefore x > 4$ and $x < 0$
lie outside parabola

$y = x$ for $x > 4$ or $x < 0$

b)



$$V = \pi \int_0^{\pi/3} \sin^2 y \, dy$$

$$= \pi \int_0^{\pi/3} \left(\frac{1}{2} - \frac{1}{2} \cos 2y \right) dy$$

$$= \pi \left[\frac{1}{2} y - \frac{1}{4} \sin 2y \right]_0^{\pi/3}$$

$$= \pi \left[\frac{\pi}{6} + \frac{\sqrt{3}}{8} \right] \text{ u}^3$$

equality will
not hold
 $p \neq q$

DO NOT NEED
TO USE
SLICES IN
EXT1

Question clearly
said y -axis not
 x -axis

Students found
incorrect
region had
to integrate
 $\sin^2 y$ to
get marks.

Suggested Solutions

Marks

Marker's Comments

c) $P(\text{2 or less rejects}) = (0.12 + 0.88)^{10}$

$P(2) + P(1) + P(0)$

$= {}^{10}C_2 (0.12)^2 (0.88)^8 + {}^{10}C_1 (0.12) (0.88)^9$

$+ {}^{10}C_0 (0.88)^{10}$

$= 0.891$

d) i) $T = 22 + Ae^{-kt}$

$t=0 \quad T=80$

$80 = 22 + Ae^0$

$\therefore A = 58 \quad \therefore T = 22 + 58e^{-kt}$

When $t=10 \quad T=60$

$\frac{38}{58} = e^{-10k}$

$-10k = \ln \frac{19}{29}$

$k = \frac{1}{10} \ln \frac{29}{19} \quad \text{OR} \quad -\frac{1}{10} \ln \frac{19}{29}$

(ii) $T = 22 + 58e^{-\frac{1}{10} \ln \frac{29}{19} t}$

$T = 30$

$30 = 22 + 58e^{-\frac{1}{10} \ln \frac{29}{19} t}$

$\frac{8}{58} = e^{-\frac{1}{10} \ln \frac{29}{19} t}$

$t = \frac{10 \ln \frac{29}{8}}{\ln \frac{29}{19}}$

$t = 46.848$

$\therefore$ takes 47 mins

1 for recognising correct probability required

1 for one correct term

1 for final answer.

well done.

MATHEMATICS: Question...14....
Extension 2!

Suggested Solutions

Marks

Marker's Comments

14. a) Test $n=1$

$$\begin{aligned} \text{LHS} &= \sum_{r=1}^1 r(r!) \\ &= 1 \times 1! \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= (1+1)! - 1 \\ &= 2! - 1 \\ &= 1 \end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$

$\therefore$ The result is true for $n=1$

Assume result is true for $n=k$, $k \in \mathbb{Z}^+$

$$\therefore 1(1!) + 2(2!) + \dots + k(k!) = (k+1)! - 1$$

Test for $n=k+1$

$$\text{RTP: } \sum_{r=1}^{k+1} r(r!) = (k+2)! - 1$$

$$\text{LHS} = 1(1!) + 2(2!) + \dots + k(k!) + (k+1)(k+1)!$$

$$= [(k+1)! - 1] + (k+1)(k+1)! \quad (\text{By assumption})$$

$$= (k+1)! [1 + (k+1)] - 1$$

$$= (k+1)!(k+2) - 1$$

$$= (k+2)! - 1$$

$\therefore$ If the result is true for $n=k$,

It is true for $n=k+1$

$\therefore$ The result is true by Mathematical Induction.

Note: Students should define k correctly!

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MATHEMATICS: Question 14....
Extension

Suggested Solutions	Marks	Marker's Comments
b) i) $x = e^x + e^{-x}$ $\frac{dx}{dt} = e^x + e^{-x}$ $= e^x + \frac{1}{e^x}$ $\frac{dx}{dt} = \frac{e^{2x} + 1}{e^x}$ $\frac{dt}{dx} = \frac{e^x}{e^{2x} + 1}$ $t = \int \frac{e^x}{e^{2x} + 1} dx$	(1) (1)	
ii) let $u = e^x$ $\frac{du}{dx} = e^x$ $t = \int \frac{1}{u^2 + 1} du$ $t = \tan^{-1} u + C$ $t = \tan^{-1}(e^x) + C$ When $t=0$, $x=0$, $\therefore \tan^{-1}(1) + C = 0$ $\therefore C = -\frac{\pi}{4}$ $\therefore t = \tan^{-1}(e^x) - \frac{\pi}{4}$ $\therefore \tan^{-1}(e^x) = t + \frac{\pi}{4}$ $e^x = \tan\left(t + \frac{\pi}{4}\right)$ $x = \ln\left[\tan\left(t + \frac{\pi}{4}\right)\right]$	(1)	

MATHEMATICS: Question 14
Extension 1

Suggested Solutions	Marks	Marker's Comments
$c) i) y_p = Vt \sin \theta - \frac{1}{2}gt^2$ $= t(V \sin \theta - \frac{1}{2}gt)$ $y_p = 0 \Rightarrow t = 0 \text{ or } \frac{2V \sin \theta}{g}$ $\text{When } t = \frac{2V \sin \theta}{g}$ $x_p = V \left(\frac{2V \sin \theta}{g} \right) \cos \theta$ $= \frac{V^2 2 \sin \theta \cos \theta}{g}$ $x = \frac{V^2 \sin 2\theta}{g}$ $ii) x_L = L - Vt \cos \theta$ $y_L = Vt \sin \theta - \frac{1}{2}gt^2 = y_p$ $\text{When } t = \frac{L}{2V \cos \theta}$ $x_L = L - \frac{V \cos \theta L}{2V \cos \theta}$ $= L - \frac{L}{2}$ $= \frac{L}{2}$ $x_p = Vt \cos \theta$ $= \frac{VL \cos \theta}{2V \cos \theta}$ $= \frac{L}{2}$ $= x_L$ $\text{Since } x_L = x_p \text{ at } t = \frac{L}{2V \cos \theta} \text{ and } y_L = y_p \text{ for all } t. \text{ Then } L \text{ and } P \text{ will collide}$ $\text{at } t = \frac{L}{2V \cos \theta}$	① ① ① ①	Note: Students should explain somehow that the trajectory of P and Q are identical but in opposite directions L units apart.

MATHEMATICS: Question...14...
Extension 2

Suggested Solutions	Marks	Marker's Comments
c) iii) when $t = \frac{L}{2V \cos \theta}$ $y_p = y_r = \frac{V \sin \theta L}{2V \cos \theta} - \frac{gL^2}{8V^2 \cos^2 \theta}$ $= L \left(\frac{\tan \theta}{2} - \frac{gL}{8V^2 \cos^2 \theta} \right)$ $y_p > 0 \rightarrow L \left(\frac{\tan \theta}{2} - \frac{gL}{8V^2 \cos^2 \theta} \right) > 0$ $\therefore 0 < L < \frac{8V^2 \cos^2 \theta \tan \theta}{2g}$ $0 < L < \frac{4V^2 \sin \theta \cos \theta}{g}$ <u>or</u> $0 < t_{\text{flight}} < t_{\text{range}}$ $0 < \frac{L}{2V \cos \theta} < \frac{2V \sin \theta}{g}$ $0 < L < \frac{4V^2 \sin \theta \cos \theta}{g}$ <u>or</u> $0 < \frac{L}{2} < x_{\text{range}}$ $0 < \frac{L}{2} < \frac{V^2 \sin 2\theta}{g}$ $0 < L < \frac{2V^2 \sin 2\theta}{g}$ $0 < L < \frac{4V^2 \sin \theta \cos \theta}{g}$	(1) (1) (1) (1) (1) (1) (1) (1) (1)	